Dijkstra’s Algorithm on Indian Cities Road Network

By Grandhe Satya Rohit, SE24UCSE150

1. Introduction

This project implements Dijkstra’s Algorithm to find the shortest path between cities in India based on road distances provided in a CSV file.

2. Algorithm Explanation

Dijkstra’s Algorithm finds the shortest path from a source node to all other nodes in a weighted graph. It works by always expanding the node with the smallest known distance using a priority queue. The algorithm updates distances to neighboring nodes if a shorter path is found.

3. Requirements

- Python 3.x
- CSV file containing city connections
- Required libraries: csv, heapq, collections (default in Python)

4. CSV Format

The CSV file must contain the following columns:
Origin,Destination,Distance

Example:
Hyderabad,Bangalore,570
Bangalore,Chennai,350

5. How to Run the Project

1. Place the CSV file (india_roads.csv) in the same directory as the Python script.
2. Open terminal in the project folder.
3. Run the program using: python Dijkstra.py
4. Enter the start city when prompted.
5. Enter the destination city to view the shortest path.

6. Output

The program displays:

- Shortest distance from the source city to all other cities
- Shortest path to the chosen destination Example output:

```
Enter start city: Hyderabad

Shortest distances from Hyderabad :

Hyderabad -> Agra = 1822.0 km
Hyderabad -> Delhi = 1582.0 km
Hyderabad -> Lucknow = 2130.0 km
Hyderabad -> Kanpur = 2099.0 km
Hyderabad -> Ahmedabad = 1220.0 km
Hyderabad -> Mumbai = 708.0 km
Hyderabad -> Pune = 560.0 km
Hyderabad -> Jaipur = 1751.0 km
Hyderabad -> Udaipur = 1475.0 km
Hyderabad -> Bengaluru = 576.0 km
Hyderabad -> Hyderabad = 0 km
Hyderabad -> Chennai = 626.0 km
Hyderabad -> Goa = 674.0 km
Hyderabad -> Bhubaneswar = 1059.0 km
Hyderabad -> Kolkata = 1489.0 km
Hyderabad -> Vishakhapatnam = 618.0 km
Hyderabad -> Patna = 1890.0 km
Hyderabad -> Kochi = 1123.0 km
Hyderabad -> Thiruvananthapuram = 1329.0 km
Hyderabad -> Varanasi = 2146.0 km

Enter destination city to see path: Varanasi

Shortest path:
Hyderabad -> Vishakhapatnam -> Bhubaneswar -> Patna -> Varanasi
```

7. Conclusion

This project demonstrates how graph search algorithms like Dijkstra's Algorithm can be applied to real-world problems such as route optimization and navigation systems.